

Actividad 5-20: Capturas - Web APIs

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Web APIs - Introduction

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A Web API is a developer's dream.

- It can extend the functionality of the browser
- It can greatly simplify complex functions
- It can provide easy syntax to complex code

What is Web API?

API stands for **A**pplication **P**rogramming **I**nterface.

A Web API is an application programming interface for the Web.

A Browser API can extend the functionality of a web browser.

A Server API can extend the functionality of a web server.

Browser APIs

All browsers have a set of built-in Web APIs to support complex operations, and to help accessing data.

Fetch API:

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JavaScript Fetch API

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The Fetch API interface allows web browser to make HTTP requests to web servers.

😊 No need for XMLHttpRequest anymore.

Browser Support

`fetch()` is an **ES6 feature**.

ES6 is fully supported in all modern browsers since June 2017:

				
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

A Fetch API Example

Web Geolocation API:

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Web Geolocation API

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Locate the User's Position

The HTML Geolocation API is used to get the geographical position of a user.

Since this can compromise privacy, the position is not available unless the user approves it.

Try It

Note

Geolocation is most accurate for devices with GPS, like smartphones.

Browser Support

The Geolocation API is supported in all browsers:

				
Yes	Yes	Yes	Yes	Yes

Web History API

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Web History API

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The Web History API provides easy methods to access the windows.history object.

The window.history object contains the URLs (Web Sites) visited by the user.

The Web History API is supported in all browsers:

				
Yes	Yes	Yes	Yes	Yes

The History back() Method

The back() method loads the previous URL in the windows.history list.

It is the same as clicking the "back arrow" in your browser.

Example

```

<script>
function goBack() {
  window.history.back();
}
</script>

```

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Pointer Events API

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The Pointer Events API is a modern web standard that provides a unified input model for handling various pointing devices, such as a mouse, pen/stylus, and touch (finger).

It simplifies development by consolidating separate mouse and touch event models into a single, hardware-agnostic system.

Note

For the modern web, pointer events is **the recommended approach** to create interactive interfaces that provide a consistent experience for all users, regardless of hardware.

Event Types

Pointer event names are **similar to mouse events**.

Just **replace** "mouse" with "pointer":

- **pointerdown**

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The Web Storage API is a simple syntax for storing and retrieving data in the browser. It is very easy to use:

Example

```
localStorage.setItem("name", "John Doe");
localStorage.getItem("name");
```

Try it Yourself »

The Web Storage API is supported in all browsers:

				
Chrome	IE/Edge	Firefox	Safari	Opera
Yes	Yes	Yes	Yes	Yes

The localStorage Object

The localStorage object provides access to a local storage for a particular Web Site. It allows you to store, read, add, modify, and delete data items for that

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Constraint Validation DOM Methods

Property	Description
checkValidity()	Returns true if an input element contains valid data.
setCustomValidity()	Sets the validationMessage property of an input element.

If an input field contains invalid data, display a message:

The checkValidity() Method

```
<input id="id1" type="number" min="100" max="300" required>
<button onClick="myFunction()">OK</button>

<p id="demo"></p>

<script>
function myFunction() {
  const inpObj = document.getElementById("id1");
  if (!inpObj.checkValidity()) {
    document.getElementById("demo").innerHTML = inpObj.validationMessage;
  }
}
```

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A web worker is a JavaScript running in the background, without affecting the performance of the page.

What is a Web Worker?

When executing scripts in an HTML page, the page becomes unresponsive until the script is finished.

A web worker is a JavaScript that runs in the background, independently of other scripts, without affecting the performance of the page. You can continue to do whatever you want: clicking, selecting things, etc., while the web worker runs in the background.

Browser Support

The numbers in the table specify the first browser versions that fully support Web Workers:

				
Chrome 4	IE 10	Firefox 3.5	Safari 4	Opera 11.5
Jan 2010	Sep 2012	Jun 2009	Jun 2009	Jun 2011

Web Workers Example